

Digital Data Acquisition Techniques: Reconstruction of a 1D tracker

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Overview

- 1 Goals
- 2 Design, Demonstration and Validation
- 3 Next steps
- 4 Backup

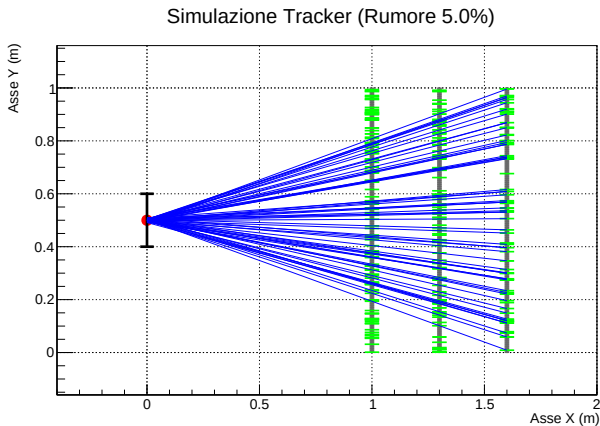


Goals & Working Assumptions

- **Goals:** Reconstruction of the trajectories of charged particles crossing N layers of a one-dimensional tracker.
- **Setup:**
 - N equally spaced layers equipped with boolean sensors distributed continuously along each layer.
 - Point-like source, with emission angles generated from specific probability distributions (e.g. Uniform and Gaussian).
 - Each layer has an acceptance of $[0, 1]$.



Experimental Set-up



Simulation of 100 charged particles crossing 3 detector layers, each made of 400 pixels











Example of the on-disk Data Format

```

00000000: caccaaaa 00000003 3f800000 3b23d70a
00000010: 3e99999a 3f800000 00000000 40a00000
00000020: 00000064 00000001 aaa00000 65400022
00000030: 10152481 61114540 0b434112 82644a22
00000040: 41a06400 2842e40c 00075620 0cc90441
00000050: 40c50012 2a082104 200210cc 00001003
00000060: aaa00001 001401aa a0021425 20485204
00000070: a2420135 08880802 680c0052 08138803
00000080: e2b84002 08020008 9010e200 03850001
00000090: 2060c044 000004a0 aaa00002 a1105140
000000a0: 11808008 22002520 41021088 91040280
000000b0: 41006800 8a851009 e0800000 00001a24
000000c0: 0000a000 188020b1 07228000 00000820
000000d0: defac0ca be411fb4 3d162642 be949d3d
000000e0: beb55434 3dac5f2e bd970405 bc47e92b
000000f0: 3e522b38 bd826ba8 3e9efe20 be615338
00000100: 3eb4e5b8 be807610 3dd70159 3dbb2df9
00000110: be2967be 3ec28b5c be9b1ec1 3bf382d0
00000120: bec3afd1 3d494d17 be852497 be83c0dd
00000130: 3ddb0719 bed6515e 3ea56851 bee7de1e
00000140: be9fc296 becd0234 3cc237da bcc4246d
00000150: bd28dc5d be96812f 3db2b038 3ec0d2f3
00000160: 3e8b8b9e 3e615c3a be9df6e0 be1198b8
00000170: 3e979ed2 3e899d06 3e683865 be5b1e6e
00000180: 3edce955 bec30380 3c64c8a1 3e4547b0
00000190: becaee23 be6f7ab6 3ec7fe6f bd20fe12
000001a0: 3e691fd4 3e9f465e beb9592d 3eb2b666
000001b0: bdcdded1 bee7e9dd be49f6ed 3e84e51c
000001c0: 3d984b65 bed19bd6 3dd23d36 be9074f7
000001d0: 3e9ed70a bdeb4606 bec090af 3daf5ff5
000001e0: be4e9e43 beahda6b bc666792 3dc39ce6
000001f0: 3e48a9a7 3ecdb42d bebc668e be1f697c
00000200: bdfcb2f6 bece9f1a 3c880cd5 3e3e287a
00000210: be095f46 bebf6a11 be45f17f bd89a3db
00000220: beec204c bed39652 3eb77fcb 3ea54363
00000230: 3ecd877a 3d15f026 beb76884 3e8a5976
00000240: 3ede2175 be192627 3ebdc68f 3e93940b
00000250: 3eab8768 becce748 3e833285 be859ee2
00000260: 3e46802a

```



CPU Track Reconstruction: χ^2 minimisation

- Iterating over the first layer, the algorithm computes the theoretical projections compatible with the detector acceptance.
- It then uses a binary search to select, on the last layer, only the contiguous portion of valid hits.
- The distance between the intermediate hits and the theoretical trajectory generated by the seed is scanned.
- A least-squares fit is performed on the final coordinates. The track is validated and stored only if the χ^2 , weighted on the pixel resolution, falls within the threshold.



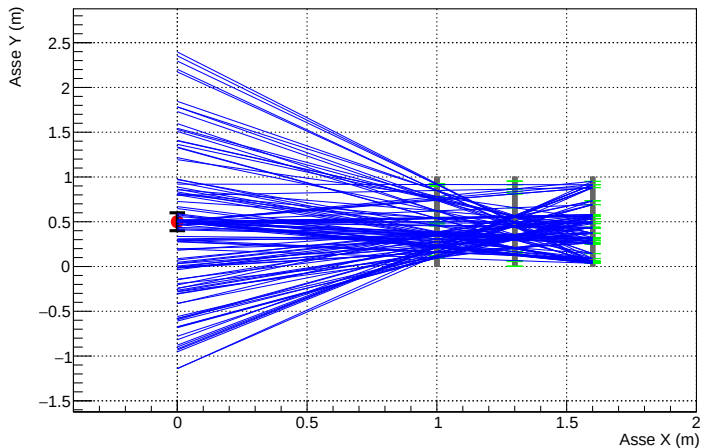
GPU Track Reconstruction: χ^2 minimisation

- 2D Thread Mapping: the kernel assigns each single thread to an exclusive combination of one first-layer hit and one last-layer hit (using 16×16 thread blocks).
- Geometric Early-Exit.
- The GPU performs a sequential linear search over the intermediate layers to avoid thread divergence.
- Storage of the tracks in shared memory with AtomicAdd.
- The data are written to VRAM buffers allocated once, persistently, with no allocation/deallocation per event.



Example:

1. Tutte le tracce ricostruite, nessun taglio su q (rumore 1%, 124 tracce)



Raw tracks found (they are the same for both CPU and GPU) without any cut on the track q. N true tracks = 30; pixel length = 0.0025 m; Noise 1%;



Track reconstruction: ghost-track removal

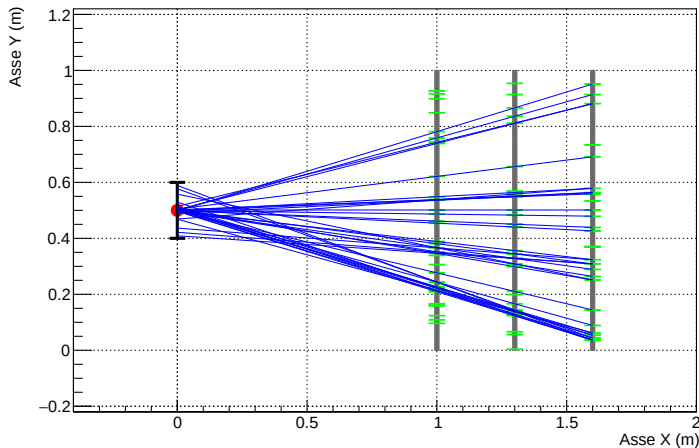
The `track_finder()` function can reconstruct the same physical particle multiple times, with slightly different parameters, when a noise hit close to the true one still passes the χ^2 cut.

- `RimuoviCloni()` identifies duplicates with two combined criteria: (m,q) parameters compatible within 3σ (the same theoretical sigmas propagated from the geometry) and at least one hit shared between the two reconstructions.
- The identified clones are merged into a single track: parameters = cluster average, χ^2 = the best among the members.



Example:

3. Rimozione delle tracce clone (rumore 1%, 35 tracce)

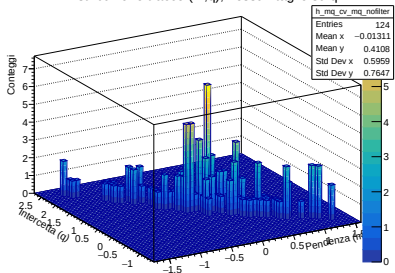


Removal of clone tracks from the raw tracks, with selection of tracks having $q \in [0.4, 0.6]$. N true tracks = 30; pixel length = 0.0025 m; Noise 1%;



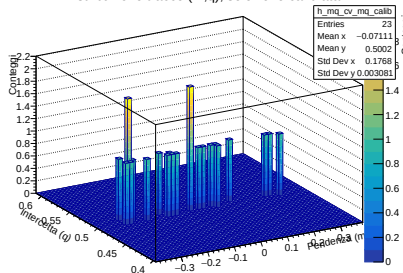
Example:

Distribuzione tracce (m,q), nessun taglio su q



N true tracks = 30; pixel length = 0.0025
m; Noise 1%;

Distribuzione tracce (m,q), selezione calibrata



N true tracks = 30; pixel length = 0.0025
m; Noise 1%;



Track reconstruction: source position

Why not a classic gaussian fit?

- The hit positions are quantised and produce a peak made of discrete lines, not a continuous gaussian.
- A least-squares fit on this structure is unstable: it can collapse onto a single tooth, with $\sigma \rightarrow 0$.



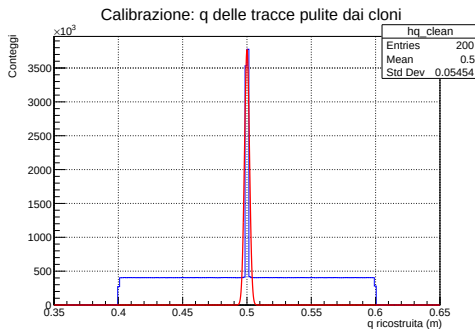
Solution

A deterministic two-stage estimate, with no non-linear minimisation.

- Stage 1: adaptive bin aggregation until the FWHM (full width at half maximum) reaches a physically credible value with respect to the theoretical sigma expected from the detector geometry $\implies$ obtain σ .
- Stage 2: iterative sigma-clipping with a fixed-width window, which only recentres the mean (μ).



Source reconstruction

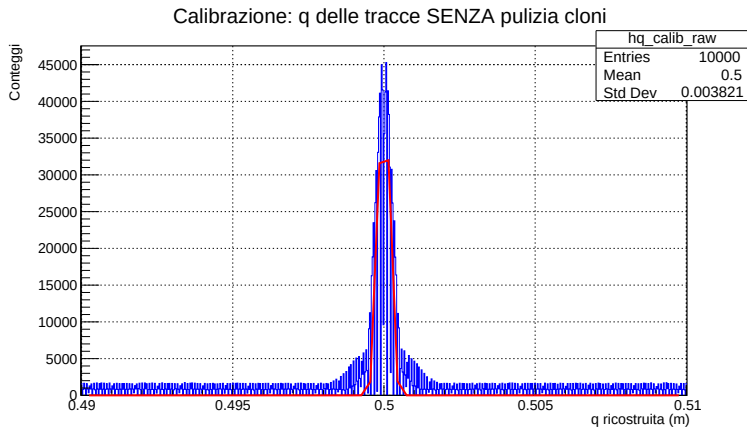


100000 calibration events were used for each N-tracks value, and efficiency and purity were then evaluated.



Fun fact

If clones are not removed and the graphical binning is increased, one notices...



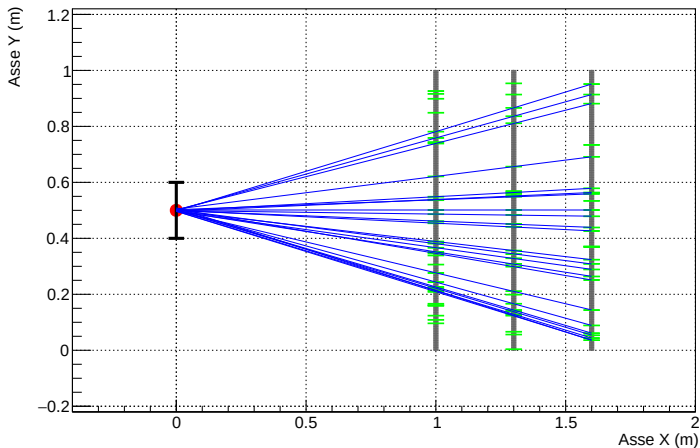
Why the comb structure?

- The detector hits are not continuous positions: each layer discretises them to the centre of the hit pixel.
- The reconstructed intercept q is a fixed-coefficient linear combination of the positions on the three layers (least-squares fit): being linear, it maps a lattice of discrete inputs into a discrete lattice of outputs.



Example:

4. Selezione calibrata sull'intercetta $\mu_q \pm 3\sigma_q$ (rumore 1%, 23 tracce)



Tracks remaining after calibration. N true tracks = 30; pixel length = 0.0025 m;
Noise 1%;



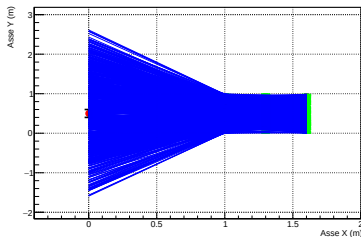
Noise

- Noise is added pixel by pixel: each still-off pixel has a fixed probability of turning on anyway, with an independent Bernoulli trial.
- Noise never switches off a true hit and does not alter pixels already lit by a particle.
- It is applied separately for each layer: no correlation between the noise of different layers.



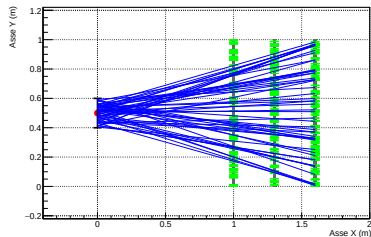
Example:

1. Tutte le tracce ricostruite, nessun taglio su q (rumore 3%, 804 tracce)



N true tracks = 0; pixel length = 0.00025
m; Noise 3%;

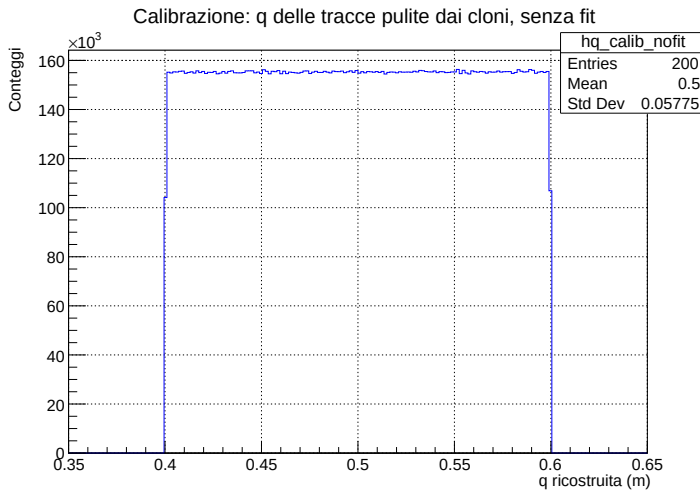
3. Rimozione delle tracce clone (rumore 3%, 55 tracce)



N true tracks = 0; pixel length = 0.00025
m; Noise 3%;



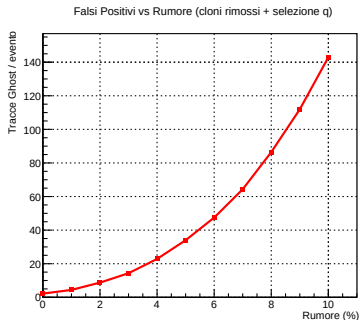
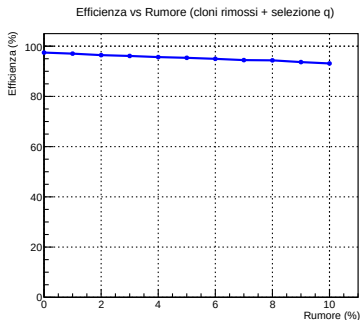
Example:



100k runs with N true tracks = 0; pixel length = 0.00025 m; Noise 3%;



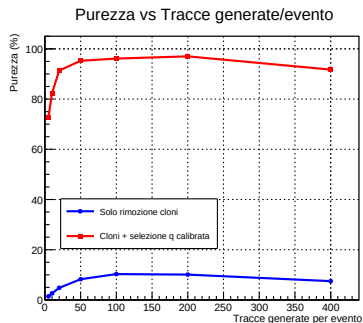
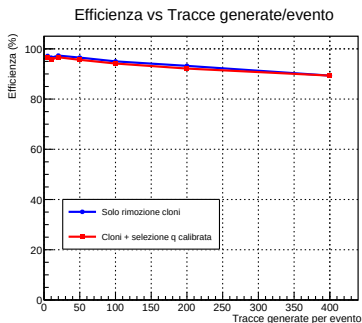
Efficiency and purity vs noise



100000 calibration events were used for each sampled noise percentage, and efficiency and purity were then evaluated.



Efficiency and purity vs N tracks

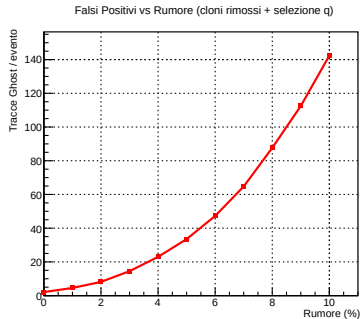
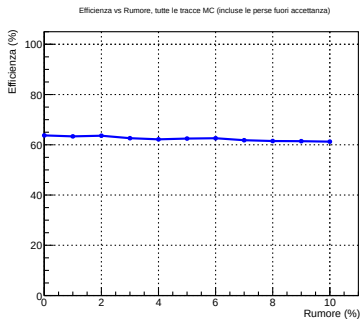


100000 calibration events were used for each N-tracks value, and efficiency and purity were then evaluated.



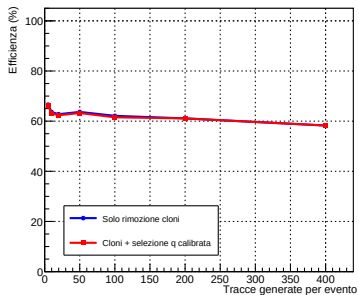
WARNING

in the efficiency and purity computation I did not account for all the tracks that exited the detector!

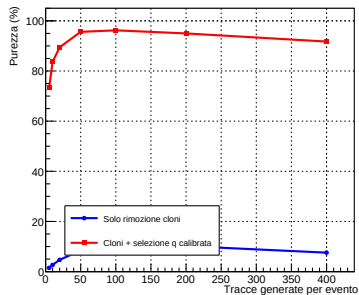


WARNING

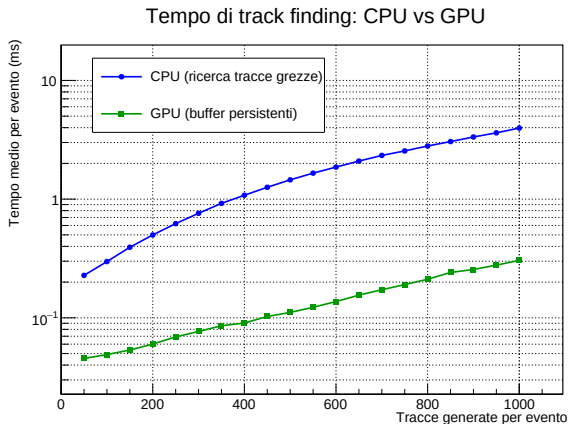
Efficienza vs Tracce generate/evento, tutte le tracce MC



Purezza vs Tracce generate/evento



Execution times



Both curves measure the raw track search only. The GPU time uses the persistent-memory engine (GpuSingleEventFinder): there remains an irreducible fixed cost per event (kernel launch + PCIe transfers) $O(1) \mu s$.



Conclusions and Next Steps

● Achievements:

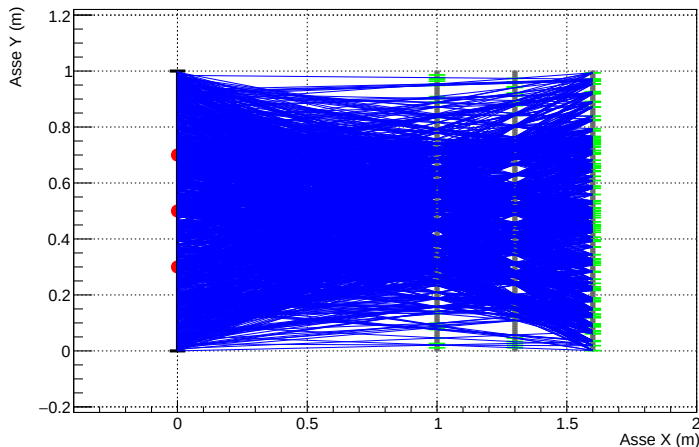
- Implementation of a 1D tracker.
- Efficient and validated event format.
- Track reconstruction with χ^2 minimisation and reconstruction of the source position.
- Optimisation of GPU performance.

● Next Steps:

- 2D/3D tracker.
- Layer misalignment.
- Introduction of dead space between pixels (non-continuous pixels).
- Track recognition with the Hough transform on FPGA, GPU-FPGA timing comparison, and comparison between the combinatorial method and the Hough transform.



2. Selezione dell'intercetta nel range della sorgente [q_min, q_max] (rumore 3%, 1356 tracce)

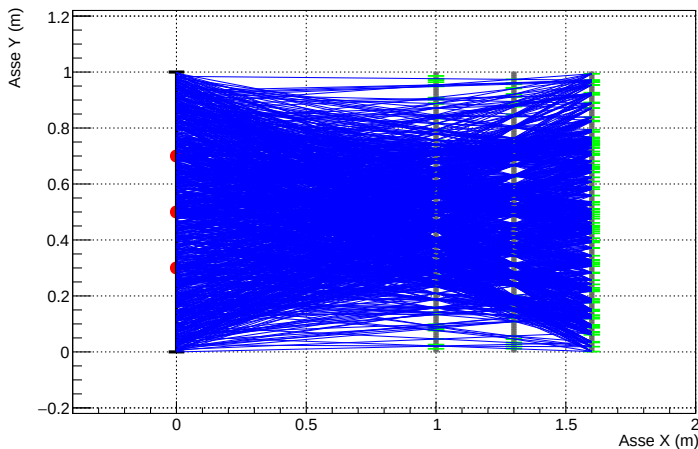


true particles = 5 / source; Pixel length = 0.0025 m; Noise = 3%



Tracker with multiple sources

3. Rimozione delle tracce clone (rumore 3%, 950 tracce)

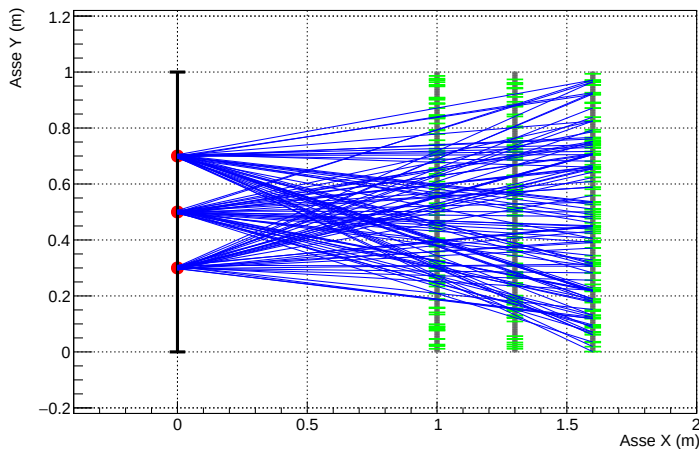


true particles = 5 / source; Pixel length = 0.0025 m; Noise = 3%



Tracker with multiple sources

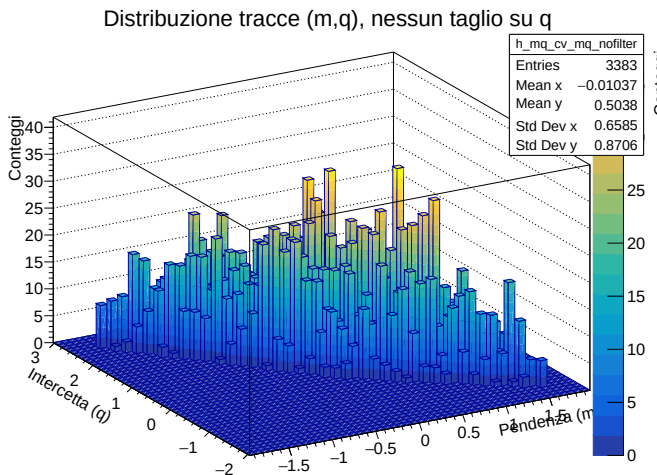
4. Selezione calibrata sull'intercetta ($\mu_q \pm 3\sigma_q$) (rumore 3%, 112 tracce)



true particles = 5 / source; Pixel length = 0.0025 m; Noise = 3%



Tracker with multiple sources

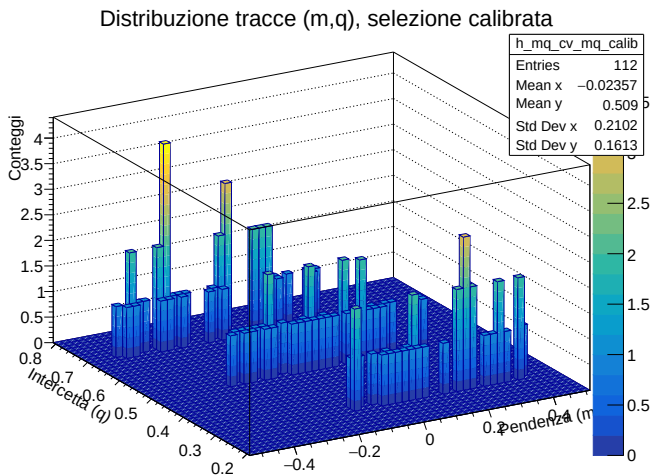


N true

particles = 5 / source; Pixel length = 0.0025 m; Noise = 3%



Tracker with multiple sources

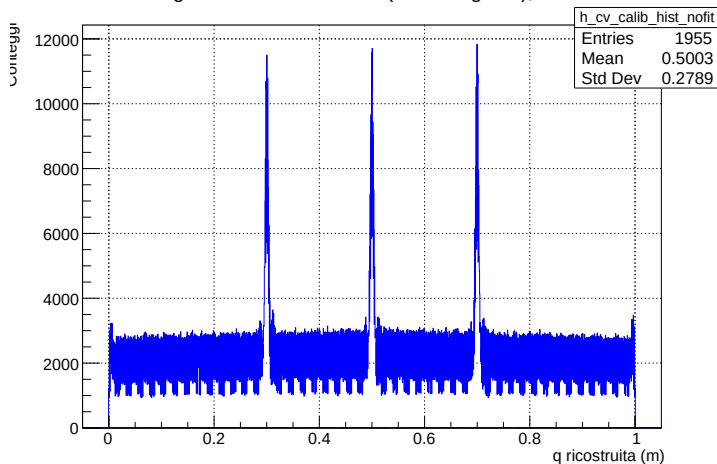


N true

particles = 5 / source; Pixel length = 0.0025 m; Noise = 3%



Istogramma di calibrazione (multi-sorgente), senza fit

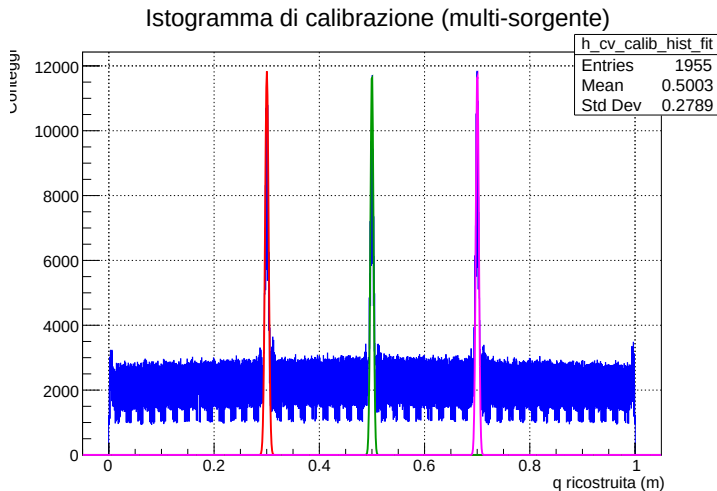


Calibration with 100k events. N true particles = 5 / source; Pixel length = 0.0025 m

Noise = 3%



Tracker with multiple sources

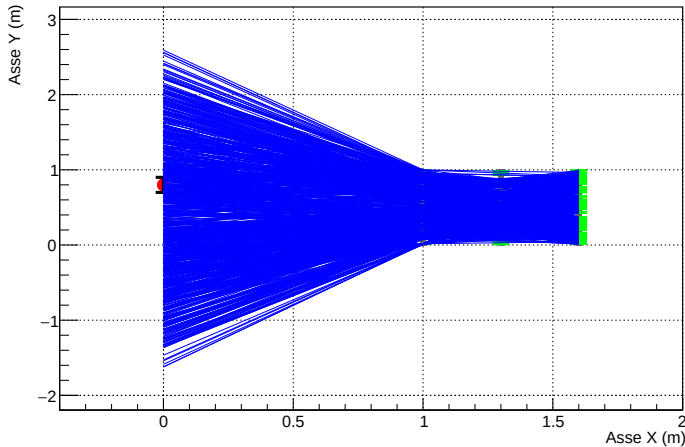


Calibration with 100k events. N true particles = 5 / source; Pixel length = 0.0025 m

Noise = 3%



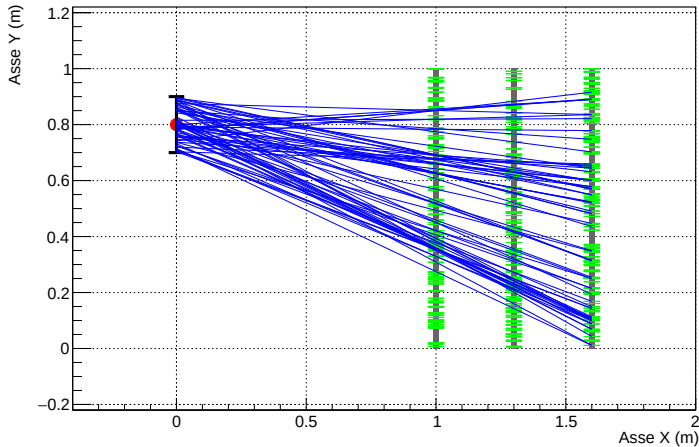
1. Tutte le tracce ricostruite, nessun taglio su q (rumore 3%, 843 tracce)



true particles = 10; Pixel length = 0.00025 m; Noise = 3%



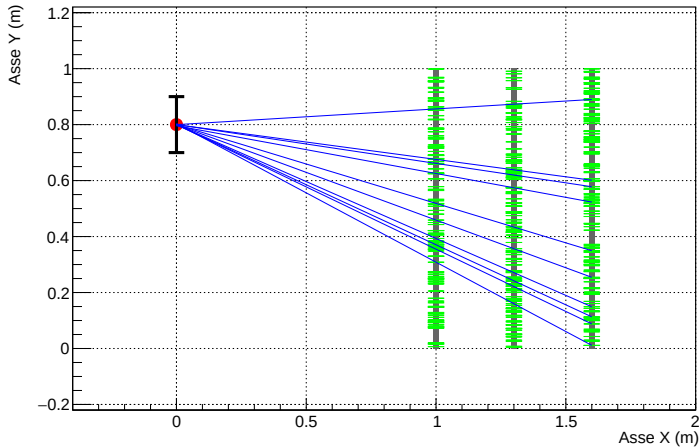
3. Rimozione delle tracce clone (rumore 3%, 67 tracce)



true particles = 10; Pixel length = 0.00025 m; Noise = 3%



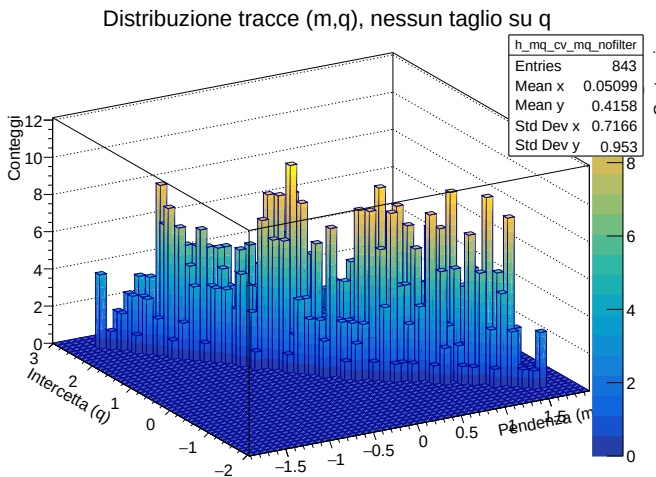
4. Selezione calibrata sull'intercetta ($\mu_q \pm 3\sigma_q$) (rumore 3%, 10 tracce)



true particles = 10; Pixel length = 0.00025 m; Noise = 3%



Displaced source

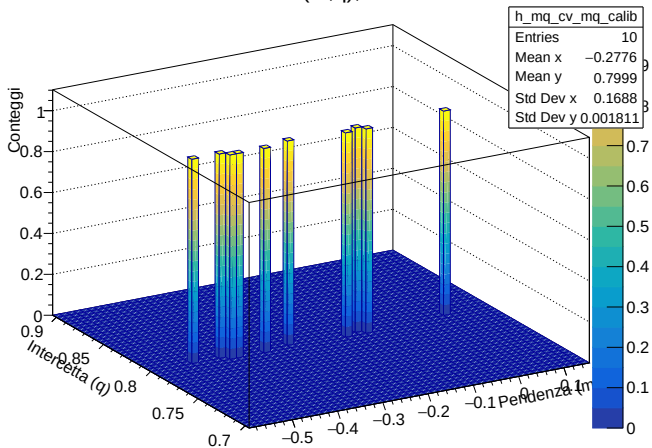


Calibration



with 100k events. N true particles = 10; Pixel length = 0.00025 m; Noise = 3%

Distribuzione tracce (m,q), selezione calibrata

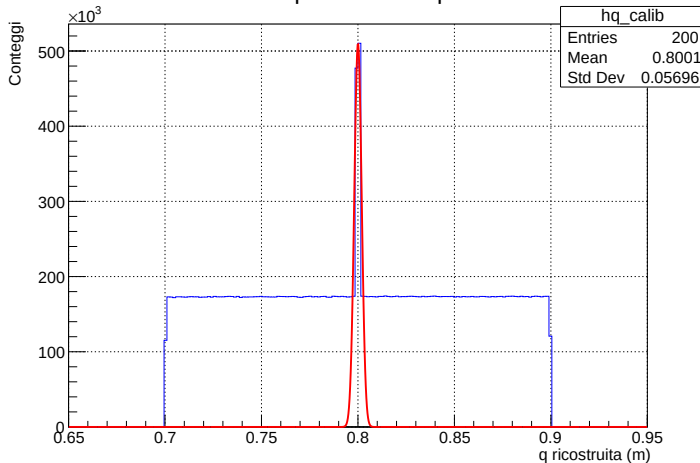


Calibration



with 100k events. N true particles = 10; Pixel length = 0.00025 m; Noise = 3%

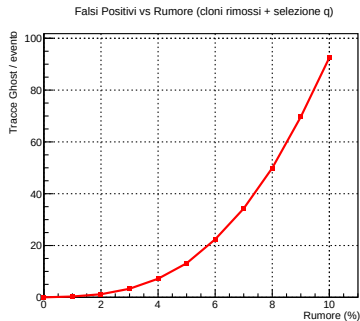
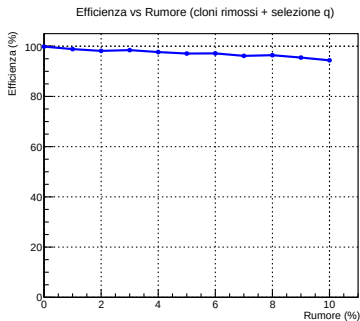
Calibrazione: q delle tracce pulite dai cloni



Calibration with 100k events. N true particles = 10; Pixel length = 0.00025 m; Noise

$$= 3\%$$

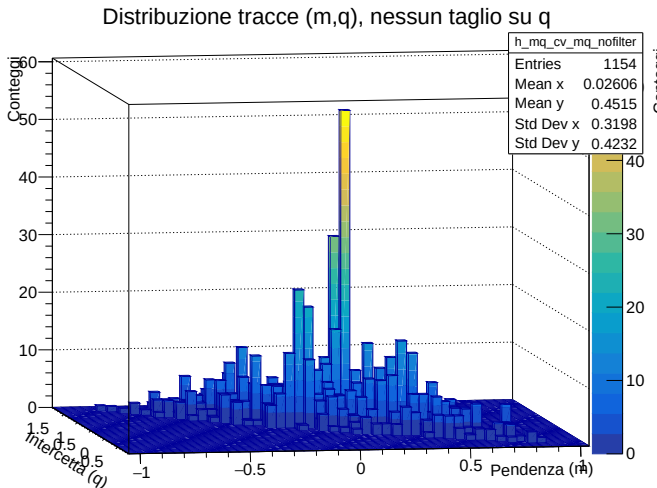

Displaced source



Calibration with 20k events per point. N true particles = 10; Pixel length = 0.00025
 m; Noise = 3%



Gaussian Generation

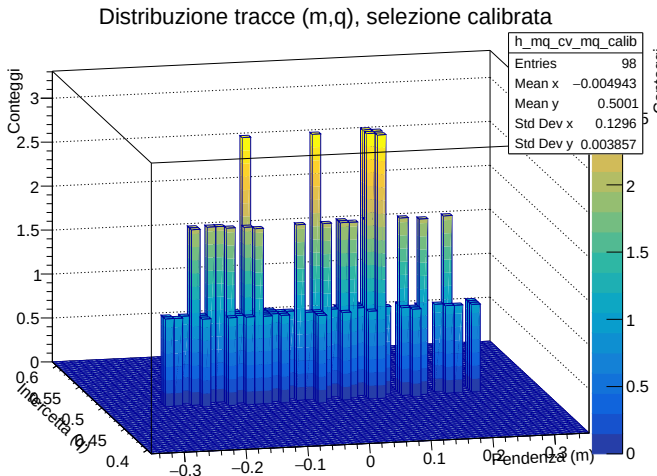


Gaussian

generation of the particle incidence angle. N true particles = 100; Pixel length = 0.0025 m; Noise = 0%



Gaussian Generation



Gaussian

generation of the particle incidence angle. N true particles = 100; Pixel length = 0.0025 m; Noise = 0%

